

Otooru Baba

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Professional Summary

Full-Stack Software Engineer with production experience across **Java**, **Python**, **JavaScript**, and **TypeScript** — building, testing, and maintaining scalable systems end-to-end. Skilled in **data structures and algorithms**, **large-scale system design**, and **RESTful API development** aligned with SOLID principles and software engineering best practices. Proven ability to triage and debug complex product issues, participate in design and code reviews, and deliver high-quality, testable code under real-world constraints. Passionate about building technology that scales to millions of users and contributes to meaningful, community-impacting products.

Technical Skills

Languages:	Java, Python, JavaScript (ES6+), TypeScript, SQL, Bash
Full-Stack:	React.js, Next.js, Angular, Redux, Tailwind CSS, HTML5, CSS3, Spring Boot, FastAPI, Node.js
System Design:	RESTful APIs, Microservices, JWT Authentication, Redis Caching, Distributed Systems, Large-Scale Pipelines
DSA:	Arrays, Trees, Graphs, Hash Maps, Dynamic Programming, Sorting & Search Algorithms
Code Health:	JUnit, Postman, Swagger, API Validation, Debugging, Root Cause Analysis, Code Review
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Tools & Cloud:	Git & GitHub, Docker, Docker Compose, GCP, AWS, IntelliJ IDEA, VS Code
AI / ML:	TensorFlow, PyTorch, Keras, FAISS, RAG, NLP, CNN, Generative AI, Sentence Transformers

Experience

Web Development & Generative AI Trainee

Sep 2025 – Dec 2025

Novacode

Remote — 10-Week Intensive Program

- Wrote and shipped full-stack product code across 5+ project cycles using **Java** backends and **JavaScript/React** frontends, adhering to style guidelines, modularity standards, and efficiency requirements throughout development.
- Participated in **design reviews** with peers and stakeholders, evaluating architectural trade-offs between available technologies and contributing to decisions that improved system testability and long-term maintainability.
- Conducted structured **code reviews** across team submissions, identifying and resolving **20+ issues** spanning accuracy, code style, edge-case handling, and performance — providing actionable feedback aligned with engineering best practices.
- Triageed and debugged **product and integration issues** end-to-end — analyzing root causes across backend services, frontend rendering failures, and API contract mismatches — and documented resolution steps to prevent recurrence.

Projects

Online Coding Judge System | Java, Spring Boot, React, TypeScript, PostgreSQL, Redis, Docker, JWT

2026

- Designed and developed a **large-scale full-stack** coding platform — architecting backend execution services in **Java/Spring Boot** and building the frontend editor with **React** and Monaco Editor, applying **data structure and algorithm** design principles to the core evaluation engine.
- Triageed and resolved **system-level performance issues** including race conditions in concurrent submission handling and cache invalidation bugs; optimized **PostgreSQL** schema and **Redis** caching layer, reducing P95 response latency by **~40%**.
- Enforced testability through **JUnit** unit and integration test suites and authored complete **Swagger** API documentation; containerized all services with **Docker Compose** and implemented structured logging for end-to-end observability.

AI Shopping Assistant – Chatbot | Python, FastAPI, React, TypeScript, Redis, FAISS, Sentence Transformers, Llama 3.1 2025

- Architected and shipped a production-grade **full-stack AI application** — **Python/FastAPI** backend with a **React/TypeScript** frontend — integrating a **RAG pipeline** (FAISS vector search + transformer embeddings) for context-aware product recommendations across **500+ product** datasets.
- Debugged and resolved critical **performance issues** including embedding latency spikes, session memory leaks, and LLM hallucination at query boundaries through systematic root cause analysis and targeted refactoring across distributed service layers.

- Designed modular, versioned **REST APIs** for search, authentication, and recommendation; applied clean architecture and separation of concerns to maximize testability, code health, and future extensibility.

Brain Tumor Detection System | *Python, TensorFlow, Keras, OpenCV, CNN* 2024

- Developed a **large-scale data pipeline** for MRI image preprocessing, augmentation, and CNN inference — applying **algorithm optimization** techniques (batch normalization, efficient tensor operations) to achieve **95%+ classification accuracy** across four tumor categories.
- Triaged and debugged model performance degradation (overfitting, class imbalance, memory overflow) through systematic **data analysis and visualization**; implemented cross-validation and iterative experimentation to resolve each failure class.

Medicinal Plant Identification Platform | *Python, TensorFlow, Streamlit, CNN, REST API, HTML, CSS* 2023

- Built and deployed an **end-to-end full-stack application** — CNN inference backend exposed as a REST API with a **Streamlit** frontend (HTML/CSS) — enabling real-time classification across **30+ plant species**; iterated on product features based on stakeholder feedback and user testing results.

Education

Kalasalingam Academy of Research and Education 2021 – 2025
Bachelor of Technology, Computer Science & Engineering CGPA: 8.3 / 10.0

Certifications

AWS Solutions Architecture Job Simulation – Forage 2026
Web Development & Generative AI – Novacode 2025
Cloud Infrastructure – Oracle 2025
Linear Regression with NumPy – Coursera 2024